

Mathematics for Machine Learning, Visually and Practically

B02 - Exact 100-page production blueprint

Audience: Learners with basic school algebra

Page architecture

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Chapter requirements

Chapter 1: Functions Graphs and Modeling

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

Chapter 2: Vectors and Geometry

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Chapter 3: Matrices and Linear Systems

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Chapter 4: Linear Transformations and Eigen Ideas

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Chapter 5: Probability Foundations

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Chapter requirements - continued

Chapter 6: Statistics Sampling and Uncertainty

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Chapter 7: Derivatives and Gradients

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Chapter 8: Optimization and Gradient Descent

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Chapter 9: Information Theory and Numerical Stability

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Chapter 10: Connecting Mathematics to Models

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Capstone

Derive and implement linear regression with gradient descent using only Python and NumPy.

Production and QA

Author original English content. Verify code, formulas, solutions, citations, diagrams, accessibility, and objective alignment. Render and reflow until the final book contains exactly 100 content pages without blank pages, repeated filler, oversized headings, or artificial spacing. Deliver source Markdown, plain text, PDF, answer guidance, and a QA report.